Design and Simulation of Sb_2Se_3-Based Asymmetric Resonant Optical Limiters Using Transfer Matrix Method

Keywords: Optical limiter; Asymmetric resonant cavity; Phase change material

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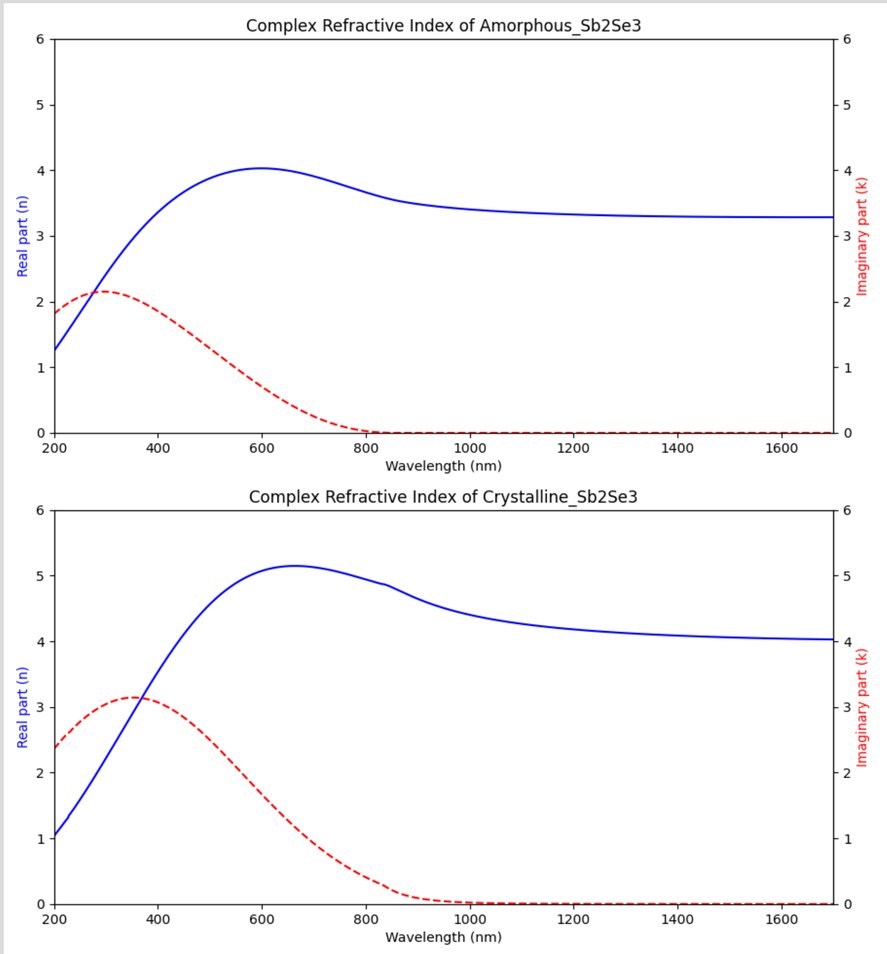
1. Introduction

Sb_2Se_3 is a non-volatile phase change material (PCM) with very low absorption in the IR spectrum[1]. Its crystallization temperature is 186°C, comparable to GST (180 °C), but is lower than other non-volatile PCMs such as GSST or Sb_2S_3[2]. Sb_2Se_3 has been demonstrated to have high phase transition endurance in the order of 10^7 cycles [2]. These properties make it attractive as a reconfigurable waveguide[3]. But its usage in another photonic device, **optical limiter**, hasn't been explored much [4].

Here, we proposed an optical limiter design utilizing asymmetrical resonant cavities comprised of Sb_2Se_3 as central layer.

2. Theory

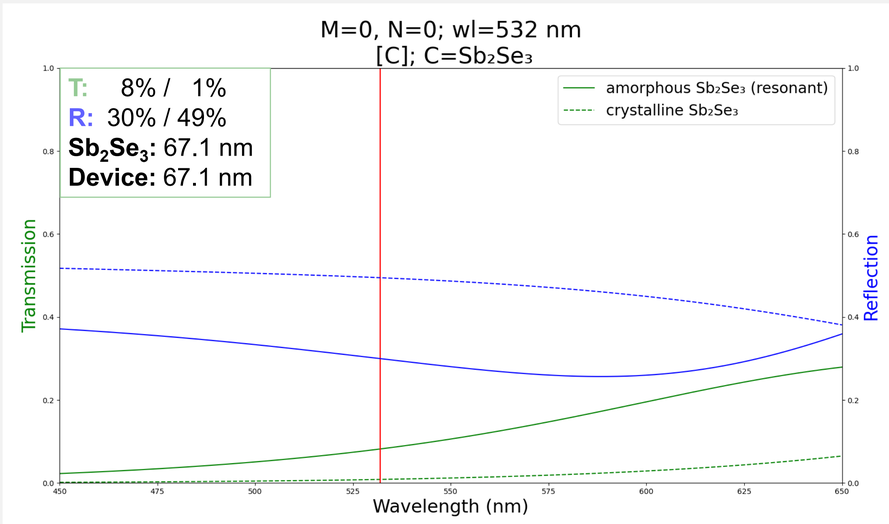
- Anti-reflection can be achieved by quarter-wave optical thickness.
- Resonant cavity allows light to cycle through the central PCM layer(s) multiple times, increasing the total absorptivity.
- We use **Asymmetrical resonant cavity** with different number of unit cell on each side, it's possible to restore resonance even with high *k index* material [5].

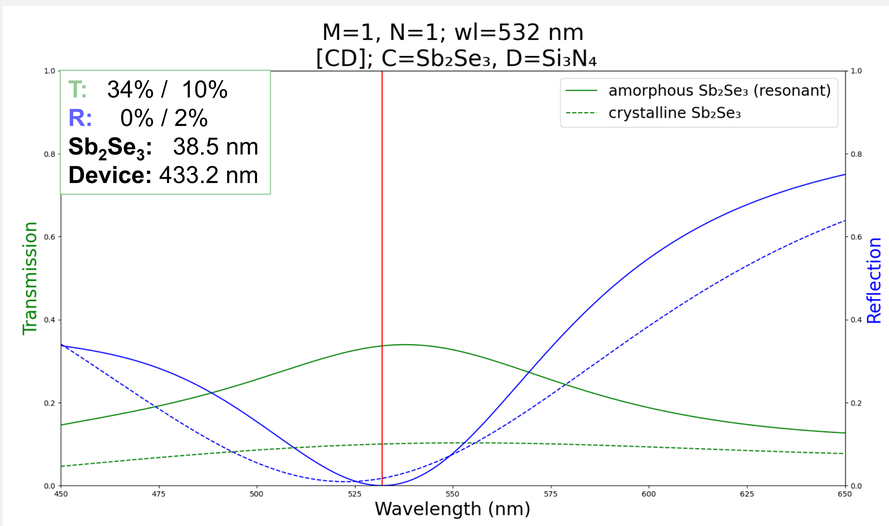


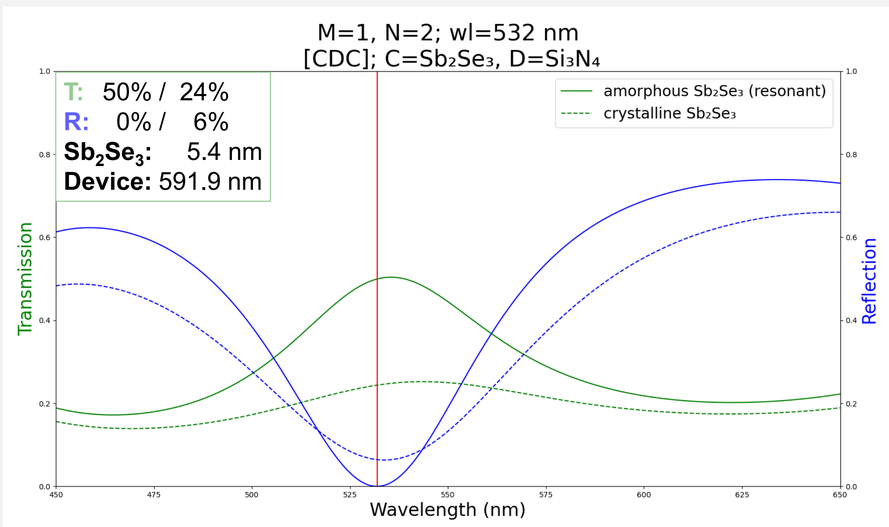
Refractive indices of amorphous (top) and crystalline (bottom) **Sb_2Se_3.** Data is taken from [6].

3. Result

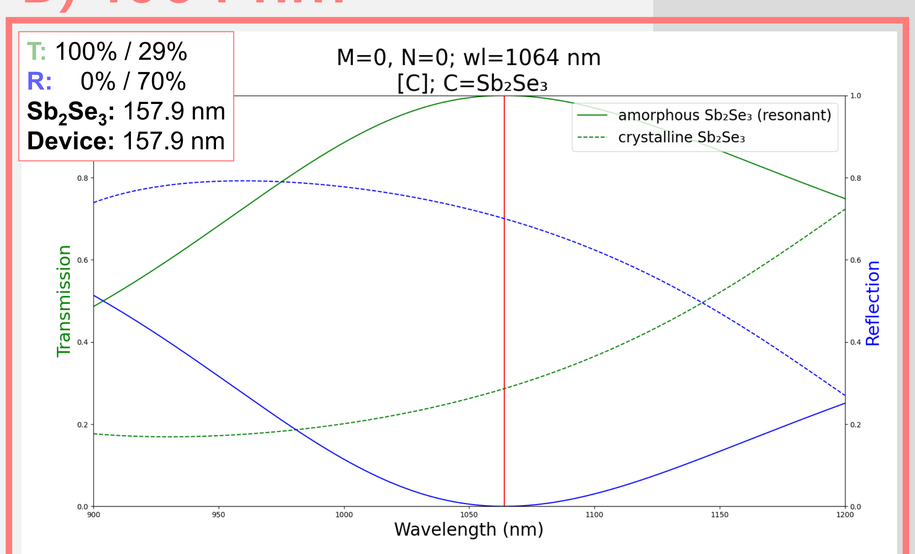
A) 532 nm

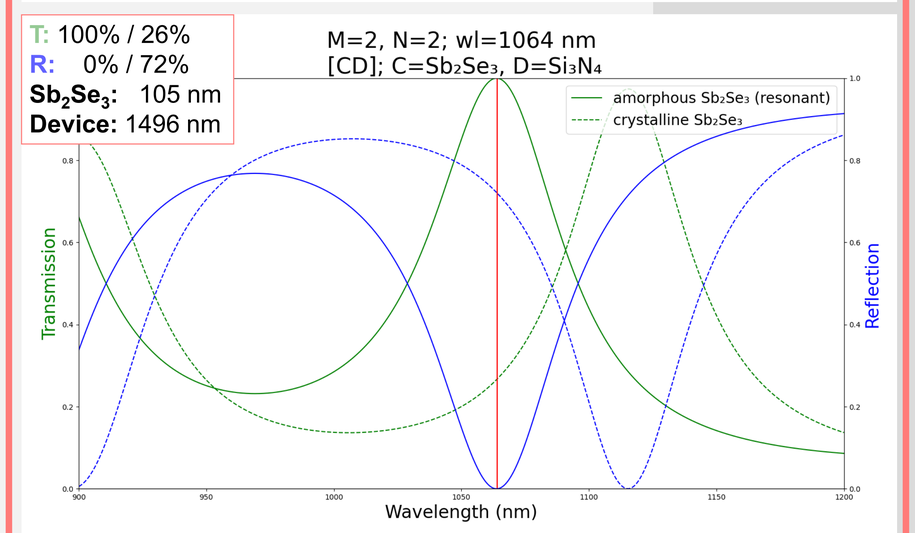


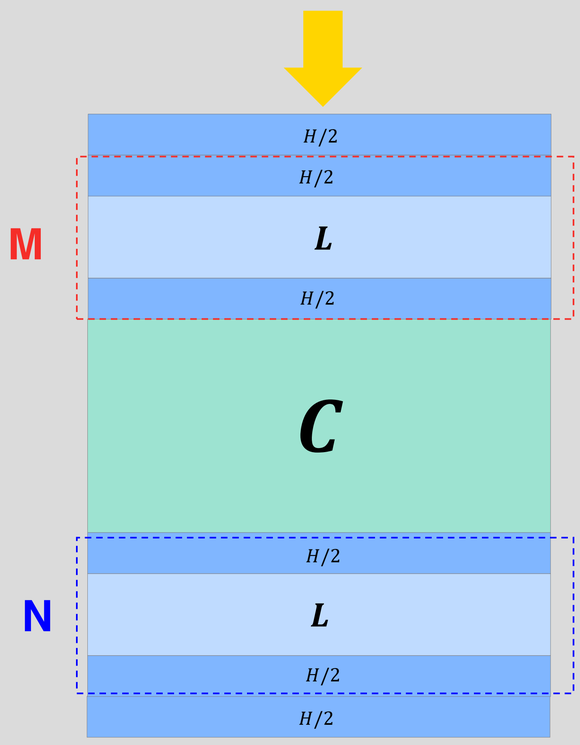




B) 1064 nm







Resonant cavity diagram. H is high refractive index value material, Si_3N_4 in this case. **L** is low-value material SiO_2. And **C** is the central layer, Sb_2Se_3 in this work.
M and **N** are the number of unit cell in the incident and substrate side, respectively.

4. Conclusion

A) 532 nm

To archive perfect antireflection property, asymmetrical design is required. Symmetrical-unit-cell design has the best transmission mode contrast ratio, while asymmetrical design have the best Δ mode contrast.

B) 1064 nm

A half-wave optical thickness film with perfect antireflection property can be achieved. While to archive better mode contrast in a cavity design, the device itself needs to be thicker, (e.g., almost ten times thicker than the original).

References

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